Who's in the data?

366 patients from a single 1998 study in Turkey (UCI Dermatology dataset). All six diseases share redness and scaling. Percentages describe this sample, not real-world prevalence: this is not a diagnostic tool.

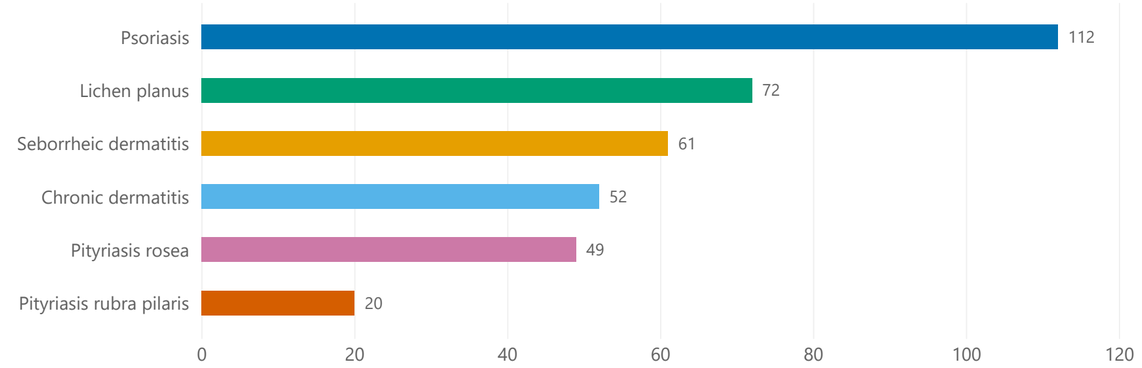
Patients
366

Median Age
35.0

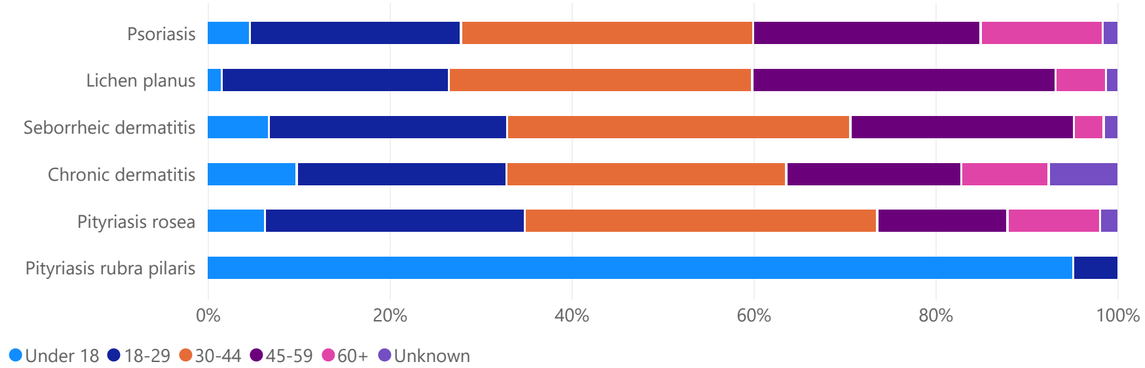
Family History %
12.6%

Unknown Ages
9

Patients by disease



Age bands by disease (share of patients)



Disease profile

Disease	Patients	Median Age	Family History %	Unknown Ages
Psoriasis	112	39.5	28.6%	2
Lichen planus	72	40.0	1.4%	1
Seborrheic dermatitis	61	34.0	4.9%	1
Chronic dermatitis	52	35.5	0.0%	4
Pityriasis rosea	49	35.0	0.0%	1
Pityriasis rubra pilaris	20	9.5	50.0%	
Total	**366**	**35.0**	**12.6%**	**9**

What makes each disease distinctive?

Signature strength = how much more common a finding is in this disease than in all other patients, in percentage points. Green = typical of the disease, red = rarer than elsewhere, white = no difference. Right-click a cell and choose Drill through > Disease detail to see a disease's top 5.

Finding group

All

Signature strength (pts) by finding and disease

Finding	Chronic dermatitis	Lichen planus	Pityriasis rosea	Pityriasis rubra pilaris	Psoriasis	Seborrheic dermatitis
Erythema	-3.2	-0.4	1.3	1.2	0.3	1.3
Scaling	-10.9	-0.7	2.5	2.3	3.1	2.6
Definite borders	-32.8	18.3	-2.6	-9.4	23.2	-22.0
Itching	19.6	36.7	-40.5	-24.1	-24.3	21.0
Koebner phenomenon	-45.2	41.6	49.5	-41.0	7.1	-44.6
Polygonal papules	-22.0	95.8	-21.8	-19.9	-27.2	-22.6
Follicular papules	9.7	-11.2	-10.4	96.2	-9.1	-8.9
Oral mucosal involvement	-21.3	93.1	-21.1	-19.4	-26.4	-22.0
Knee and elbow involvement	-32.1	-37.4	-36.3	67.3	69.2	-29.8
Scalp involvement	-32.5	-31.2	-32.2	2.3	74.3	-23.6
Melanin incontinence	-22.3	97.2	-22.1	-20.2	-27.6	-23.0
Eosinophils in the infiltrate	-4.4	3.0	-6.2	-12.1	-12.7	29.5
PNL infiltrate	-41.7	-44.6	-27.2	-22.0	48.8	43.6
Fibrosis of the papillary dermis	99.4	-14.9	-17.0	-15.6	-21.3	-17.7
Exocytosis	-7.3	38.4	34.9	23.5	-73.2	36.7
Acanthosis	3.2	1.7	-6.3	2.9	2.7	-4.6
Hyperkeratosis	11.8	-16.2	-15.6	33.9	21.2	-23.9
Parakeratosis	-33.1	3.3	-12.9	14.3	32.6	-19.0
Clubbing of the rete ridges	-29.6	-38.8	-36.0	-22.4	95.4	-37.4
Elongation of the rete ridges	51.8	-57.1	-53.0	-38.0	78.0	-41.3
Thinning of the suprapapillary epidermis	-32.8	-37.4	-34.7	-31.8	95.6	-34.1
Spongiform pustule	-20.1	-23.8	-22.1	-14.9	49.6	-7.2
Munro microabscess	-25.5	-25.5	-22.9	-23.1	68.9	-26.2
Focal hypergranulosis	-22.6	96.9	-22.4	-15.2	-28.0	-23.3
Disappearance of the granular layer	-29.6	-16.1	13.1	-26.9	48.3	-30.5
Vacuolisation and damage of basal layer	-22.9	98.3	-22.7	-20.8	-27.1	-23.6
Spongiosis	-26.3	7.2	58.1	25.8	-65.7	57.4
Saw-tooth appearance of retes	-22.9	98.3	-20.4	-20.8	-28.3	-23.6
Follicular horn plug	-4.8	-5.8	-6.9	94.1	-8.7	-5.2
Perifollicular parakeratosis	-6.7	-7.1	-6.6	99.7	-8.3	-4.9
Inflammatory mononuclear infiltrate	1.9	2.7	1.7	-1.5	-1.3	-3.6
Band-like infiltrate	-22.3	98.3	-24.3	-17.0	-27.7	-23.3

Which look-alike pairs need a biopsy?

A finding is sharp for a pair when its prevalence in the two diseases differs by at least the cutoff. Each dot is a pair: the share of its 10 clinical findings that are sharp (across) against the share of its 22 biopsy findings (up). Top-left = biopsy-dependent.

PSO psoriasis, SD seborrheic dermatitis, LP lichen planus, PR pityriasis rosea, CD chronic dermatitis, PRP pityriasis rubra pilaris.

Sharp gap cutoff (points)

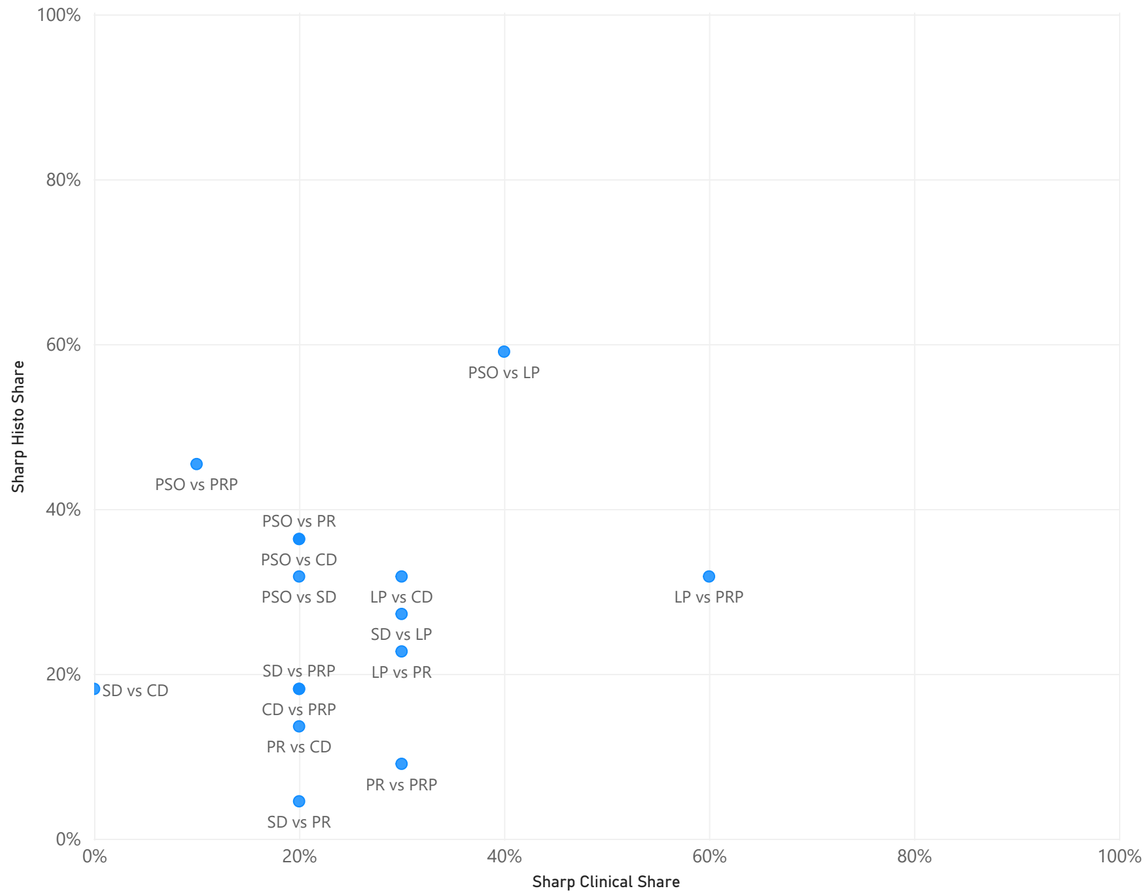
50

Biopsy-Dependent Pairs

1

At 50 points only SD vs CD needs a biopsy; slide to 80 and three psoriasis pairs join it.

Clinical vs biopsy separability, one dot per pair



Pairs ranked by biopsy lift

Pair	Sharp Clinical Share	Sharp Histo Share	Biopsy Lift (pts)	Pair Verdict
PSO vs PRP	10.0%	45.5%	35.5	Clinically separable
PSO vs LP	40.0%	59.1%	19.1	Clinically separable
SD vs CD	0.0%	18.2%	18.2	Biopsy-dependent
PSO vs CD	20.0%	36.4%	16.4	Clinically separable
PSO vs PR	20.0%	36.4%	16.4	Clinically separable
PSO vs SD	20.0%	31.8%	11.8	Clinically separable
LP vs CD	30.0%	31.8%	1.8	Clinically separable
CD vs PRP	20.0%	18.2%	-1.8	Clinically separable
SD vs PRP	20.0%	18.2%	-1.8	Clinically separable
SD vs LP	30.0%	27.3%	-2.7	Clinically separable
PR vs CD	20.0%	13.6%	-6.4	Clinically separable
LP vs PR	30.0%	22.7%	-7.3	Clinically separable
SD vs PR	20.0%	4.5%	-15.5	Clinically separable
PR vs PRP	30.0%	9.1%	-20.9	Clinically separable
LP vs PRP	60.0%	31.8%	-28.2	Clinically separable

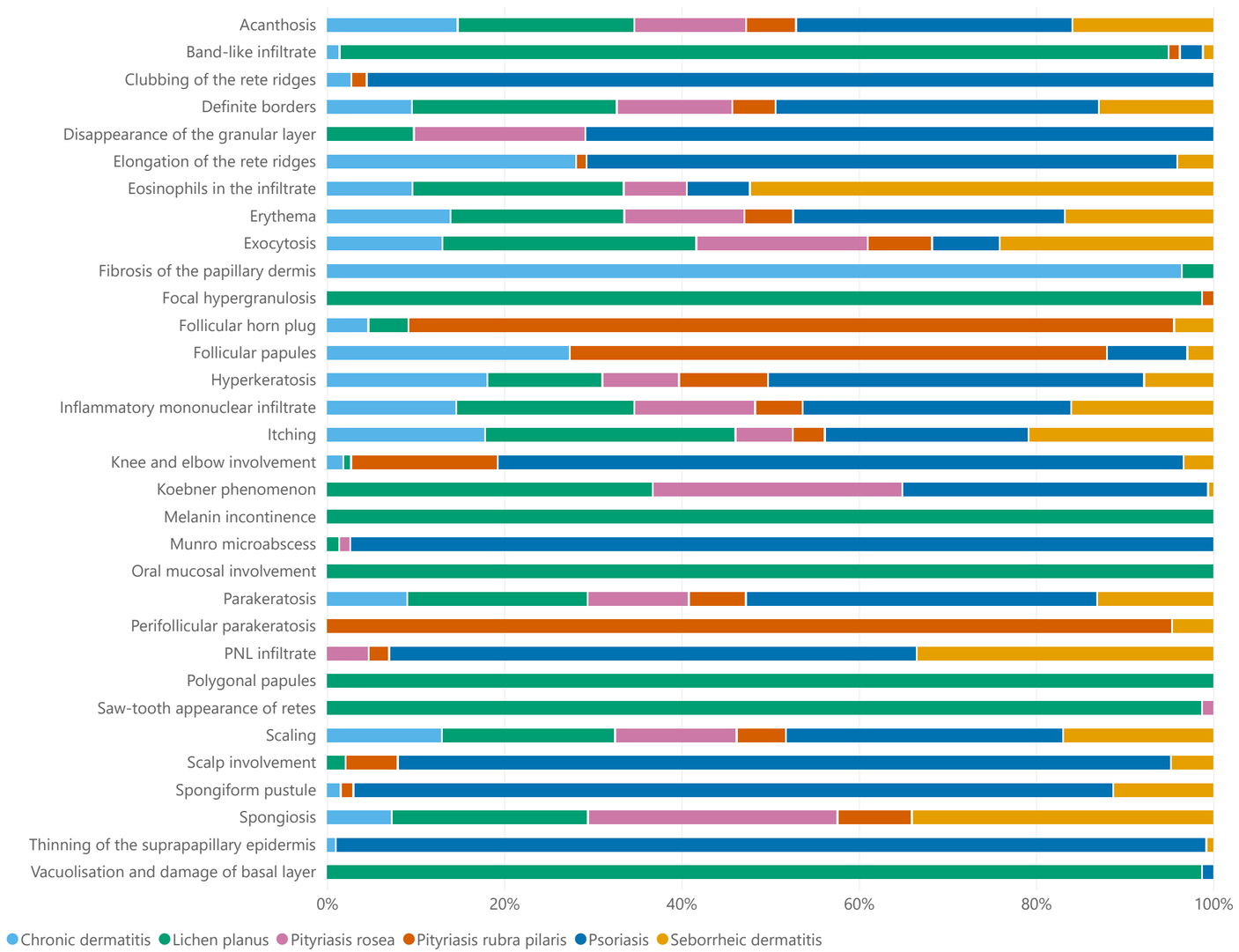
Which findings point to one disease? Who is atypical?

Bars: which diseases the patients with each finding have, counting a finding when its score is at least k. A red flag is a finding where one disease is at least 80% of those patients (with at least 10 patients). Atypical = missing at least one of the disease's top 3 signature findings.

Minimum score counted (k)

1

Which diseases have each finding (share of patients with it)



Red flags at k = 1 and k = 2 (one disease is at least 80%)

k	Finding	Points to	Share %	Patients
1	Melanin incontinence	Lichen planus	100.0	70
1	Oral mucosal involvement	Lichen planus	100.0	67
1	Polygonal papules	Lichen planus	100.0	69
1	Focal hypergranulosis	Lichen planus	98.6	71
1	Saw-tooth appearance of retes	Lichen planus	98.6	72
1	Vacuolisation and damage of basal layer	Lichen planus	98.6	72
1	Thinning of the suprapapillary epidermis	Psoriasis	98.2	110
1	Munro microabscess	Psoriasis	97.5	80
1	Fibrosis of the papillary dermis	Chronic dermatitis	96.3	54
1	Clubbing of the rete ridges	Psoriasis	95.6	114
1	Perifollicular parakeratosis	Pityriasis rubra pilaris	95.2	21
1	Band-like infiltrate	Lichen planus	93.5	77
1	Scalp involvement	Psoriasis	87.3	102
1	Follicular horn plug	Pityriasis rubra pilaris	86.4	22
1	Spongiform pustule	Psoriasis	85.7	70

Atypical patients by disease

